Data Communication Lab Assignment

Lab 1: Vertical Redundancy Check

Suppose the sender wans to send the word ***world***. In ASCII the five characters (number of characters should be variable) are coded (with even parity) as,

1110111 1101111 1110010 1101100 1100100

Find out the bits sent.
Enter the character that the receiver has received.
Show the result. Determine whether the receiver has detected the errors or not.

Lab 2: Checksum

Suppose the following block of 16 bits is to be sent using a checksum of 8 (or multiple of 8) bits.

10101001 00111001

Find out the checksum. Add the checksum to the message.
Enter the data received by the receiver. Determine whether there is any error or not.

Lab 3: CRC

Suppose the to be send by the sender is **100100** and the divisor is **1101**. Find out the CRC. Add the CRC to the bits to be send. Enter the data received by the receiver. Determine whether the receiver received the original data pr not. **You will try to enter the message data and the divisor from the input.**

Lab 4: Hamming Code

Enter the data to send. The number of input bits will be maximum 15. Find out the hamming code. Change one bit. The position of the bit should be entered from the input. Find out the position of the bit that has been sent.